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# Isolation, characterization and biological activities of phenolics from the roots bark of *Morus alba*

Min Tan [1], Xiao-Cong Liu [1], Dan Liu, Xuan-Qin Chen, Rong-Tao Li, Zhi-Jun Zhang *

*Faculty of Life Science and Technology, Kunming University of Science and Technology, Kunming, 650500, Yunnan, PR China*

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ABSTRACT

An undescribed coumarin–limonene hybrid (morusalin A, **1**), an unreported steppogenin–limonene hybrid (morusalin B, **2**), and three undescribed isoprenylated flavonoids (morusalins C–E, **3**–**5**), along with forty-one known ones (**6**–**46**), were isolated from the dried roots bark of *Morus alba*. Their structures with absolute configurations were elucidated by detailed interpretation of NMR spectroscopy, mass spectrometry, single-crystal X-ray diffraction, and ECD calculations. In addition, all these isolated compounds were assayed for their inhibitory activity against lipopolysaccharide (LPS)-induced nitric oxide (NO) production in murine macro-phage RAW 264.7 cells and for their cytotoxic activities against human colon cancer (HCT 8) and hepatoma (HepG 2) cells. Moreover, a biosynthetic pathway for the formation of compounds **1**–**6** is also proposed.

## 1. Introduction

*Morus alba* L. (Moraceae), known as white mulberry, is a species native to China and now widely cultivated in other countries (Hao et al., 2021). Different parts of the *M. alba* tree have been used as traditional medicines for thousands of years (Dong et al., 2023). The white mulberry leaves, an important food for silkworm, are used to treat hypertension, arthritis, and the fruit is a diuretic and tonic agent (Dat et al., 2010). In China, the root bark of *M. alba* (Chinese name "Sang Bai Pi") was used for its neuroprotective, anti-inflammatory, anti-tumor, antibacterial properties, and anti-oxidative activities (Hao et al., 2022; Zuo et al., 2018, 2019). Previous chemical investigations on *M. alba* led to the identification of an array of isoprenoid-substituted phenolics (Pu et al., 2024; Zheng et al., 2017), sugars (Chan et al., 2016), and alkaloids (Kim et al., 2014), Diels-Alder type adducts (Xia et al., 2019). In particular, the isoprene-substituted flavanone derivatives exhibit a variety of significant biological activities (Langeder et al., 2023; Zheng et al., 2017).

Based on these findings, it has inspired us to explore additional compounds with distinctive structures and biological activities derived from this plant. Therefore, a systematic chemical analysis to the ethanol extract of the roots bark of *M. alba* was conducted by NMR spectroscopy, mass spectrometry, single-crystal X-ray diffraction, and the biological activities of these components were evaluated. Here, it reported the isolation and structural identification of these compounds, as well as the evaluated of the anti-inflammatory effects and cytotoxicity against HCT 8 and HepG 2 cells.

## 2. Results and discussion

The dried roots bark of *M. alba* was extracted with ethanol to yield the crude extract, which was suspended in water and mixed with ethyl acetate (EtOAc) to obtain the EtOAc extract. The EtOAc extract was separated by silica gel column chromatography, ODS, reversed-phase HPLC and gel column chromatography to obtain morusalins A–E (**1**–**5**) and forty-one known ones (Fig. 1).

Compound **1** was purified as a white powder, and established to possess a molecular composition of $C_{19}H_{16}O_3$, according to the HRESIMS (*m/z* 293.1163 $[M+H]^+$, calcd for 293.1172), indicating 12 degrees of unsaturation. The IR spectrum showed absorption bands for a lactone C═O group at 1733 $cm^{-1}$, and aromatic ring (1624 and 1456 $cm^{-1}$). The $^1$H NMR spectrum (Table 1) of **1** showed distinctive resonances for seven hydrogens at low field [$\delta_H$ 7.80 (1H, s, H-5), 7.72 (1H, d, $J = 9.5$ Hz, H-4), 7.54 (1H, s, H-6′), 7.16 (1H, s, H-8′), 7.16 (1H, s, H-9′), 6.90 (1H, s, H-8), 6.30 (1H, d, $J = 9.5$ Hz, H-3)], and three methyl [$\delta_H$ 2.42 (3H, s, $H_3$-10′), 1.65 (6H, s, $H_3$-4′, $H_3$-5′)]. The $^{13}$C NMR and DEPT data (Table 1) exhibited nineteen carbon signals including nine quaternary carbons, seven methines, three methyls. The $^1$H and $^{13}$C

* Corresponding author.
*E-mail address:* zzj0121@126.com (Z.-J. Zhang).
[1] M. Tan and X.-C. Liu made equal contributions to this work.

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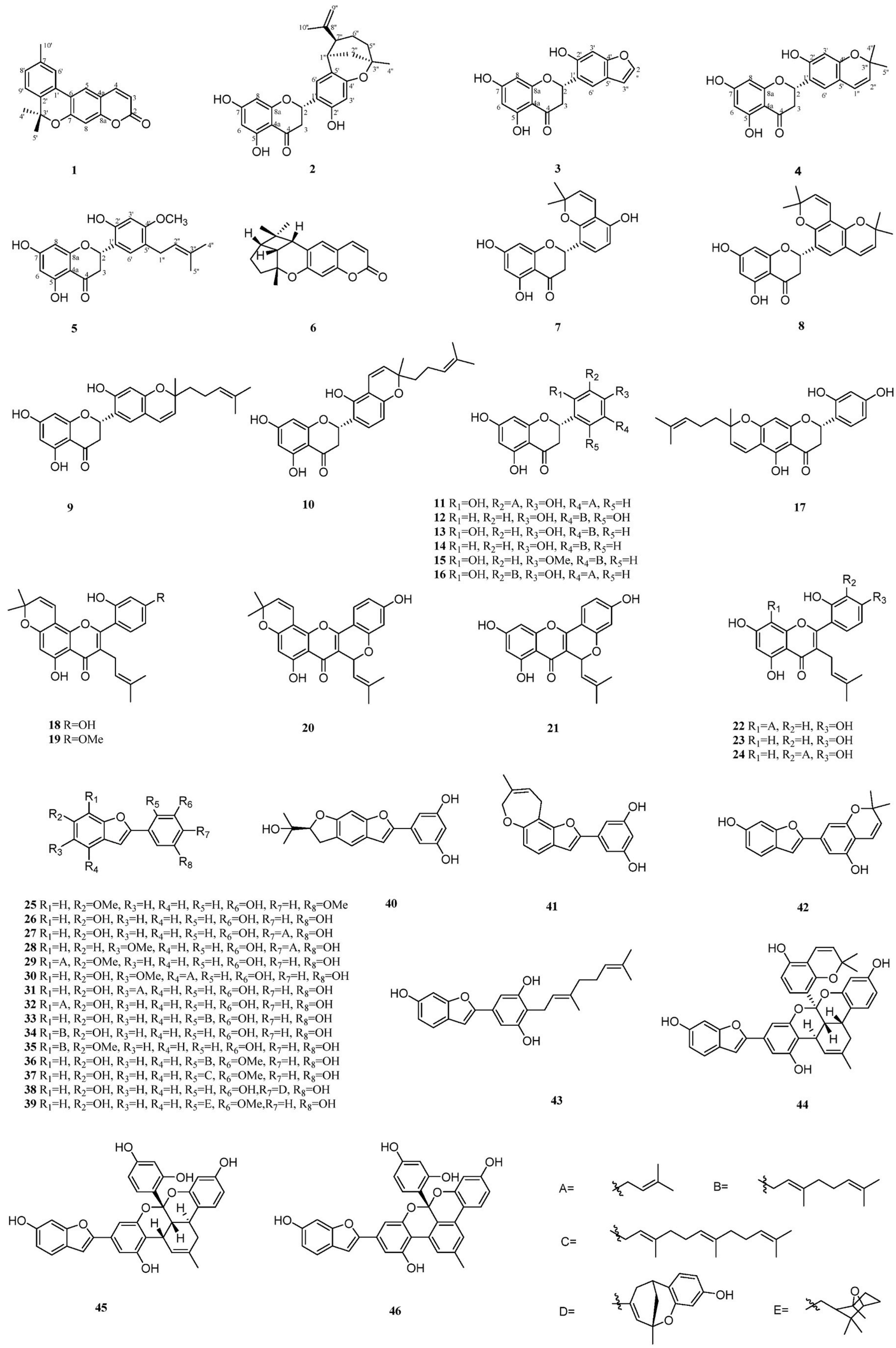


**Fig. 1.** Chemical structures of compounds **1**–**46** from *M. alba.*

**Table 1**
$^1$H-NMR (600 MHz),$^{13}$C-NMR (150 MHz) spectral data for morusalin A (**1**) in $CDCl_3$.

| position | $\delta_C$ | $\delta_H$ (*J* in Hz) | position | $\delta_C$ | $\delta_H$ (*J* in Hz) |
|---|---|---|---|---|---|
| 2 | 161.3 | | 2′ | 136.4 | |
| 3 | 113.7 | 6.30 (1H, d, 9.5) | 3′ | 79.3 | |
| 4 | 143.7 | 7.72 (1H, d, 9.5) | 4′ | 28.1 | 1.65 (3H, s) |
| 4a | 113.7 | | 5′ | 28.1 | 1.65 (3H, s) |
| 5 | 122.0 | 7.80 (1H, s) | 6′ | 122.9 | 7.54 (1H, s) |
| 6 | 120.2 | | 7′ | 137.9 | |
| 7 | 156.8 | | 8′ | 129.4 | 7.16 (1H, s) |
| 8 | 106.0 | 6.90 (1H, s) | 9′ | 123.6 | 7.16 (1H, s) |
| 8a | 155.4 | | 10′ | 21.5 | 2.42 (3H, s) |
| 1′ | 126.9 | | | | |

NMR spectra of compound **1** showed similar patterns to those of xanthyletin (Cazal et al., 2009). The first difference between them was the absence of the two alkene protons at $\delta_H$ 5.71 (1H, d, *J* = 9.6 Hz) and 6.36 (1H, d, *J* = 9.6 Hz), and reflected in the chemical shifts of C-1′ [$\delta_C$ 104.3 (d) in xanthyletin; 126.9 (s) in **1**] and C-2′ [$\delta_C$ 112.9 (d) in xanthyletin and 136.4 (s) in **1**]. The second difference between them was the presence of three additional aromatic protons at $\delta_H$ 7.54 (1H, s) and 7.16 (2H, s), and one methyl at $\delta_H$ 2.42 (3H, s) in **1**, and reflected in the chemical shifts of C-6′ ($\delta_C$ 122.9), C-7′ ($\delta_C$ 137.9), C-8′ ($\delta_C$ 129.5), C-9′ ($\delta_C$ 123.6), and C-10′ ($\delta_C$ 21.5). Based on the HMBC correlations of the H-4′ and H-5′ with C-2′ and C-3′, H-6′ with C-6, C-1′, C-2′, C-6′, and C-8′, H-9′ with C-1′, C-2′, C-3′, and C-7′, and H-10′ with C-6′, C-7′, and C-8′, the skeleton of the compound was determined. Therefore, the complete structure of compound **1** was successfully confirmed, and it had been tentatively designated as morusalin A.

Compound **2** was acquired as a white amorphous powder and the molecular formula was speculated to be $C_{25}H_{26}O_6$ based on its HR-ESI-MS (*m/z* 423.1813 [M+H] $^+$, calcd for 423.1802). The IR spectrum showed the presence of hydroxyl (3425 cm$^{-1}$) and carbonyl (1635 cm$^{-1}$) groups. The signals depicted in the $^1$H NMR (Table 2) of **2** included seven hydrogen signals at low field [$\delta_H$ 6.61 (1H, s, H-6′), 6.14 (1H, s, H-3′), 5.83 (1H, d, *J* = 2.1 Hz, H-8), 5.81 (1H, d, *J* = 2.1 Hz, H-6), and 4.61 (1H, s, H-9″a), and 4.05 (1H, s, H-9″b)], two methyl peaks in the high field [$\delta_H$ 1.69 (3H, s, $H_3$-10″) and 1.26 (3H, s, $H_3$-4″)]. After detailed analysis of its $^{13}$C NMR (Table 2) and DEPT data, 25 carbon signals were observed, including eleven quaternary carbons, seven methines, five methylenes, and two methyls. The carbon signals observed in the $^{13}$C NMR spectroscopic spectrum of **2** bore a striking resemblance to those of morflavanone A (Cao et al., 2018). The primary distinction between them was the chemical shifts of C-8″ [$\delta_C$ 73.5 (s) in morflavanone A; 147.8 (s) in **2**] and C-9″ [$\delta_C$ 27.7 (q) in morflavanone A and 111.0 (t) in **2**]. It was verified by HMBC correlations (Fig. 2) from $H_2$-9″ to C-10″, C-7″, and C-8″, $H_3$-10″ to C-7″, C-9″, and C-8’’. This inference was further confirmed by the $^1$H–$^1$H COSY correlation from H-2 to $H_2$-3, H-1″ to $H_2$-2″ and H-7″, H-7″ to $H_2$-6″, $H_2$-6″ to $H_2$-5″. The correlations observed in the ROESY spectra (Fig. 2) of H-1″/$H_3$-4″ and H-1″/$H_3$-10″, indicated that H-1″, $H_3$-4″, $H_3$-10″ was α-oriented. Additionally, the correlation between H-5b″/H-7″ and H-2b″/H-7″ indicated that a β-orientation for H-7″. The single-crystal X-ray crystallography by employing graphite-monochromated Mo Kα radiation of **2** confirmed the above deduction (Fig. 3). Crystallographic data of it have been deposited at the Cambridge Crystallographic Data Centre (CCDC) (deposition No. 2358494). The absolute configuration of **2** was determined to be 2*S*, 1″*S*, 3″*S*, 7″*R* by correlating its experimental ECD spectrum with calculated models (Fig. 4a). Therefore, compound **2** was identified as morusalin B.

**Table 2**
$^1$H-NMR (600 MHz),$^{13}$C-NMR (150 MHz) spectral data for morusalin B (**2**) in $CD_3OD$.

| position | $\delta_C$ | $\delta_H$ (*J* in Hz) | position | $\delta_C$ | $\delta_H$ (*J* in Hz) |
|---|---|---|---|---|---|
| 2 | 75.4 | 5.57 dd (10.9, 3.1) | 6′ | 129.2 | 6.61 s |
| 3 | 42.7 | 2.85 dd (17.1, 10.9) | 1″ | 36.8 | 2.89 m |
| | | 2.63 dd (17.1, 3.1) | 2″ | 38.2 | 1.86 d (12.1) |
| 4 | 198.0 | | | | 1.78 d (12.1) |
| 4a | 103.4 | | 3″ | 75.6 | |
| 5 | 165.4 | | 4″ | 29.2 | 1.26 s |
| 6 | 96.8 | 5.81 d (2.1) | 5″ | 40.5 | 1.83 m |
| 7 | 168.4 | | | | 1.52 m |
| 8 | 96.0 | 5.83 d (2.1) | 6″ | 23.8 | 1.26 m |
| 8a | 165.3 | | 7″ | 49.4 | 2.13 d (8.9) |
| 1′ | 116.5 | | 8″ | 147.8 | |
| 2′ | 154.9 | | 9″ | 111.0 | 4.61 s |
| 3′ | 102.1 | 6.14 s | | | 4.05 s |
| 4′ | 158.4 | | 10″ | 23.1 | 1.69 s |
| 5′ | 114.9 | | | | |

Compound **3** was isolated as a white powder with a molecular formula of $C_{17}H_{12}O_6$ from its HRESIMS ions at *m/z* 313.0710 [M+H]+ (calcd for $C_{17}H_{13}O_6^+$, 313.0707), implying 12 degrees of unsaturation. The IR spectrum showed the presence of a hydroxyl group (3390 and 3228 cm$^{-1}$) and a keto-carbonyl group (1640 cm$^{-1}$). Its $^1$H NMR spectrum (Table 3) showed seven methines, including two oxygenated methine proton resonance [$\delta_H$ 5.65 (1H, dd, *J* = 12.8, 2.2 Hz, H-2) and 7.50 (1H, d, *J* = 2.2 Hz, H-2″)], one methylene [$\delta_H$ 2.91 (1H, dd, *J* = 18.6, 12.8 Hz, H-3a) and 2.78 (1H, dd, *J* = 18.6, 2.2 Hz, H-3b)]. The $^{13}$C NMR and DEPT spectra (Table 3) exhibited 17 carbons, including nine quaternary carbons, seven methines, one methylene. The NMR signals at $\delta_{C/H}$ 145.3/7.50 (1H, d, *J* = 2.2 Hz, H-2″), 107.5/6.65 (1H, d, *J* = 2.2 Hz, H-3″) suggested the presence of a furan ring moiety in the structure of **3**. The above NMR spectroscopic data of **3** showed high similarity to those of cudraflavanone E (Chen et al., 2013). The only difference is that the absence of the prenyl group carbon signals ($\delta_C$ 131.2, 123.8, 26.0, 21.8, and 18.0) and the appearance of aromatic proton at $\delta_H$ 5.88 (1H, d, *J* = 2.1 Hz). The aromatic proton at $\delta_H$ 5.88 was then assigned to H-6 due to the HMBC correlations of $\delta_H$ 5.88 (H-6) is related to $\delta_C$ 168.4 (C-7), 165.5 (C-5), 103.3 (C-4a), 97.1 (C-8) (Fig. 2). Meanwhile, the furan ring was fused to C-4′ (156.9) and C-5′ (121.2), according to their HMBC correlations as shown in Fig. 2. The absolute configuration of C-2 was suggested to be *S* from the large coupling constant (*J* = 12.8 Hz) between the axial protones of H-2 and H-3 (Slade et al., 2005). In addition, the optical rotation values of the (2*S*)-flavanones [$[\alpha]_D^{25}$ = - 42.3 (*c* 1.0, MeOH) for cudraflavanone E (Chen et al., 2013); $[\alpha]_D^{20}$ = - 40 (*c* 0.2, MeOH) for sanggenol Q (Jung et al., 2015); $[\alpha]_D^{20}$ = - 54 (*c* 0.3, MeOH) for shandougenine C (Luo et al., 2014)] and (2*S*)-flavanones [$[\alpha]_D^{20}$ = + 5.3 (*c* 0.5, MeOH) for davidone B (Ma et al., 2021); $[\alpha]_D^{20}$ = + 54.0 (*c* 0.1, MeOH) for (2*R*)-kurarinol (Yang et al., 2021)] also support this inference. Besides, the absolute configuration of **3** was also determined to be 2*S* by correlating its experimental ECD spectrum with calculated models (Fig. 4b). Based on the above data analysis, the structure of **3** was confirmed and named morusalin C.

Compound **4** was obtained as a white powder, its molecular formula was determined to be $C_{20}H_{18}O_6$ based on the positive-ion HRESIMS at *m/z* 355.1177 [M+H]+(calcd for $C_{20}H_{19}O_6^+$, 355.1176). The IR absorption bands at 3432 and 1638 cm$^{-1}$ indicated the presence of hydroxyl and carbonyl group. The $^1$H NMR spectrum (Table 3) of compound **4** exhibit six hydrogen signals in the low field [$\delta_H$ 7.01 (1H, s, H-6′), 6.23 (1H, d, *J* = 9.5 Hz, H-1″), 6.18 (1H, s, H-3′), 5.85 (1H, d, *J* = 1.7 Hz, H-8), 5.81 (1H, d, *J* = 1.7 Hz, H-6), 5.53 (1H, d, *J* = 13.2 Hz, H-2), and 5.43 (1H, d, *J* = 9.5 Hz, H-2″)], along with two methyl peaks in the high field [$\delta_H$ 1.30 (6H, s, $H_3$-4″, $H_3$-5″)]. The $^{13}$C NMR and DEPT spectra (Table 3) depicted 20 carbon resonances, including ten quaternary carbons, seven methines carbon, one methylene, and two methyls. Comparison of the NMR data between compound **4** and **8** (Jung et al., 2015) revealed striking similarities, the primary distinction between them was the absence of five carbon signals ($\delta_C$ 28.2, 28.3, 77.4, 115.1, and 128.7) in **4**. The remaining 2,2-dimethyl-pyran ring was fused at C-4′ and C-5′ due to the HMBC correlations (Fig. 2) of H-1″, H-2″, H-3′ and H-6′ with C-5′, of H-1″ and H-6′ with C-4′, and of H-1″ and C-6′. The $J_{2,3}$ =

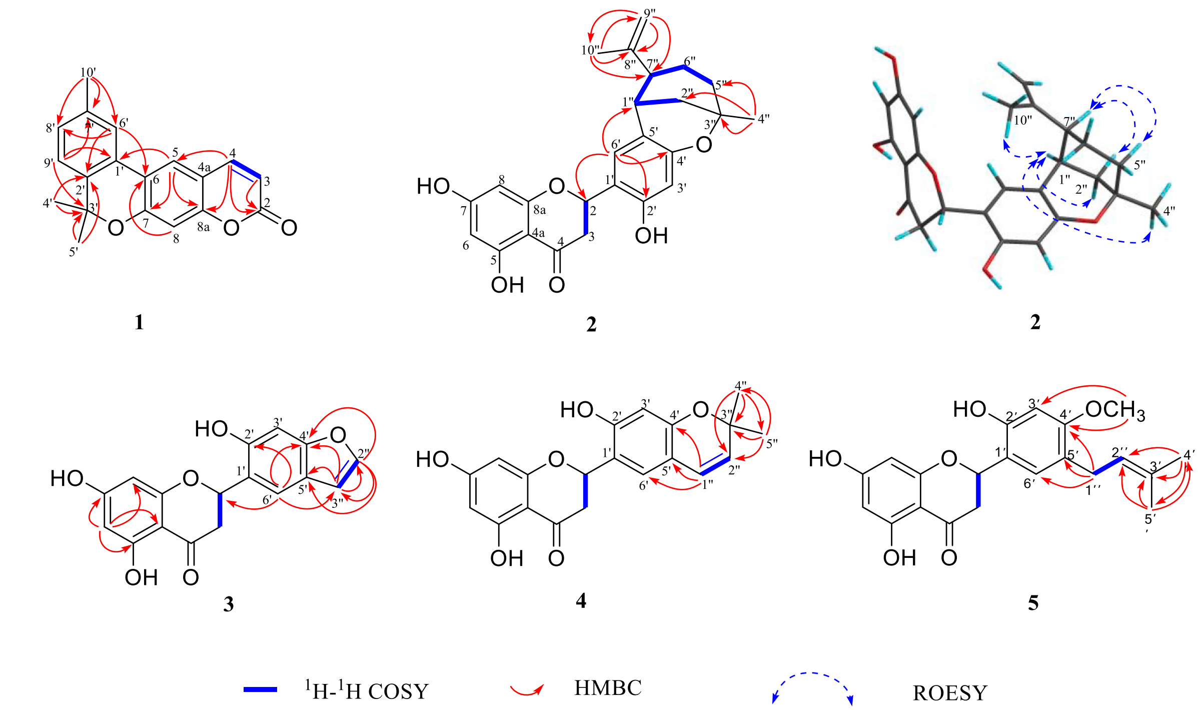


**Fig. 2.** Key HMBC, $^{1}H$–$^{1}H$ COSY and ROESY correlations of compounds **1**–**5**.

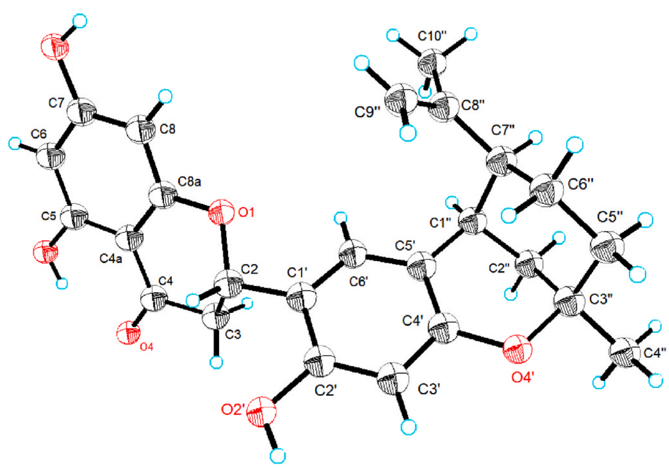

**Fig. 3.** X-ray ORTEP drawings of compound **2**.

13.2 Hz and the $[\alpha]_D^{25}$ = - 26.7 (c 0.12, MeOH) for **4**, suggesting that the absolute configurations at C-2 was *S* configuration. Furthermore, the calculated ECD spectrum of (2*S*)-**4** was consistent with the experimental one (Fig. 4c). Based on the above analysis, compound **4** was deduced and trivial named morusalin D.

Compound **5** was acquired as colorless oil, with a molecular formula determined to be $C_{21}H_{22}O_6$ from the quasi-molecular ion peak at *m/z* 371.1494 [M+H]+(calcd for 371.1489) in the HRESIMS, and IR spectra of **5** showed many resemblances to those of **4**. The $^{1}H$ NMR data (Table 3) show that compound **5** have six hydrogen signals, one methoxy, and two methyls. The $^{13}C$ NMR and DEPT spectra (Table 3) depicted 21 carbon resonances, consisting of ten quaternary carbons, six methine, and three methyls. The carbon signals observed in the $^{13}C$ NMR spectroscopic spectrum of **5** bore a striking resemblance to those of a known compound, kuwanon U (Jeong et al., 2009). The major difference between them was the absence of five carbon signals ($\delta_C$ 17.8, 26.0, 43.0, 124.3, and 132.6) in **5**, suggesting the geranyl side chain in kuwanon U was replaced by a prenyl group in **5**. This inference was further confirmed by HMBC correlations as shown in Fig. 2. The $J_{2,3}$ = 12.5 Hz and the $[\alpha]_D^{25}$ = - 29.6 (c 0.19, MeOH) for **5**, suggesting that the absolute configurations at C-2 was *S* configuration. Furthermore, the calculated ECD curve of (2*S*)-**5** closely corresponded to the experimental one as depicted in Fig. 4d. Thus, the identification of compound **5** was confirmed, and it was designated as morusalin E.

The structure of known compounds, paratrimerin F (**6**) (Dang et al., 2017), sanggenon F (**7**) (Jung et al., 2015), sanggenol O (**8**) (Jung et al., 2015), kuwanon F (**9**) (Takasugi et al., 1979), sanggenon N (**10**) (Jung et al., 2015), sanggenol Q (**11**) (Jung et al., 2015), sanggenol A (**12**) (Jung et al., 2015), kuwanon E (**13**) (Jeong et al., 2009), propolin H (**14**) (Weng et al., 2007), 4′-methoxykuwanon E (**15**) (Smejkal et al., 2010), sanggenol P (**16**) (Zheng et al., 2012), sanggenol L (**17**) (Jung et al., 2015), morusin (**18**) (Ryu et al., 2008), 5-hydroxy-2-(2-hydroxy-4-methoxyphenyl)-8,8-dimethyl-3-(3-methyl-2-buten-1-yl)-4*H*, 8*H*-benzo[1,2-*b*:3,4-*b*′]dipyran-4-one (**19**) (Tseng et al., 2010), cyclomorusin (**20**) (Ryu et al., 2008), cyclocommunol (**21**) (Lan et al., 2013), kuwanon C (**22**) (Ryu et al., 2008), albanin A (**23**) (Lin et al., 2016), kuwanon T (**24**) (Chang et al., 2019), morusalfuran E (**25**) (Ha et al., 2016), moracin M (**26**) (Jeong et al., 2009), moracin C (**27**) (Yang et al., 2012), sanggenofuran B (**28**) (Shi et al., 2007), 3′,5′-dihydroxy-6-methoxy-7-prenyl-2-arylbenzofuran (**29**) (Yang et al., 2011), moracin T (**30**) (Kapche et al., 2009), moracin N (**31**) (Yang et al., 2012), moracin S (**32**) (Kapche et al., 2009), albafuran A (**33**) (Jeong et al., 2009), mulberrofuran L (**34**) (Shi et al., 2001), mulberrofuran B (**35**) (Chang et al., 2019), mulberrofuran A (**36**) (Chang et al., 2019), morusalfuran C (**37**) (Ha et al., 2016), mulberrofuran H (**38**) (Paudel et al., 2019), 2-[5-hydroxy-3-methoxy-2-[(1,3,3-trimethyl-7-oxabicyclo [2.2.1] hept-2-yl)methyl]phenyl]-6-benzofuranol (**39**) (Zhang et al., 2014), moracin O (**40**) (Zoofishan et al., 2019), moracin G (**41**) (Takasugi et al., 1979), moracin D (**42**) (Yang et al., 2012), albafuran B (**43**) (Christensen and Lam, 1989), mulberrofuran K (**44**) (Shim et al., 2018), mulberrofuran G (**45**) (Shim et al., 2018), and mulberrofuran P (**46**) (Fukai et al., 1985) were established by comparison with literature values.

A possible biosynthetic pathway is proposed for morusalins A–E

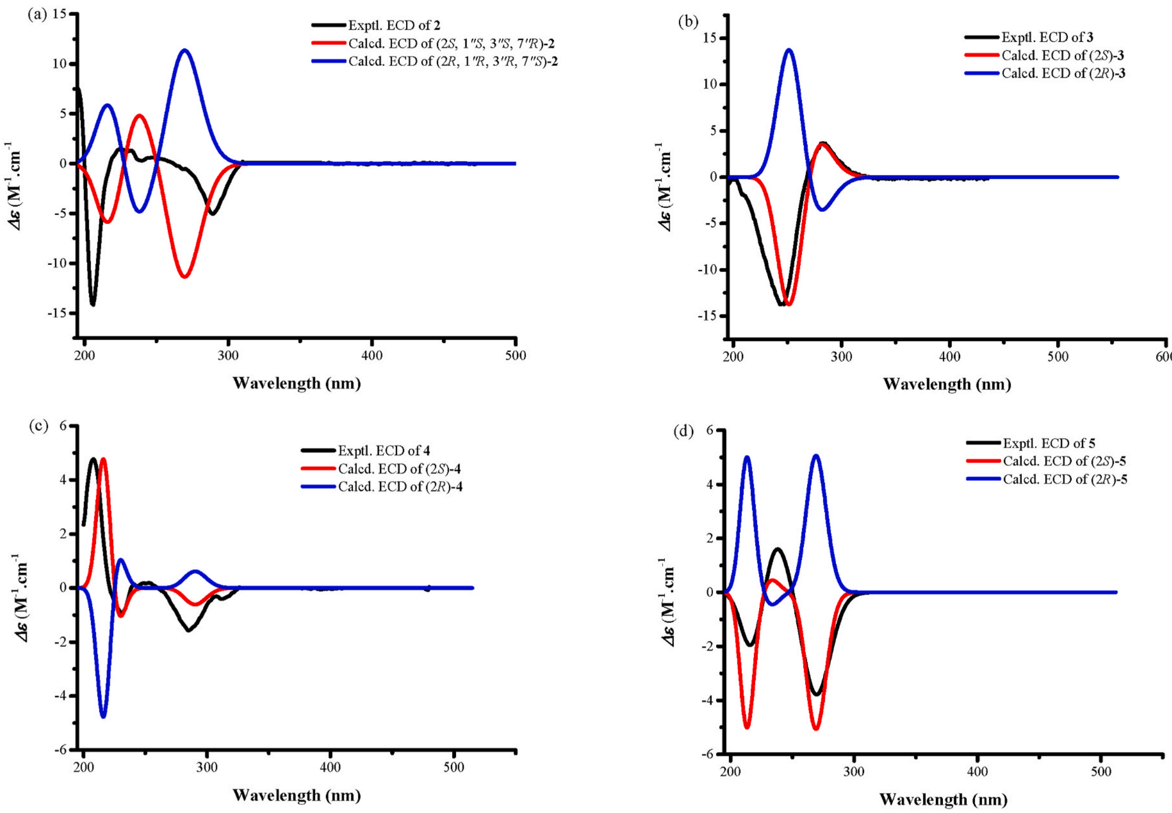


**Fig. 4.** Experimental and calculated ECD spectra of compound **2**–**5**.

**Table 3**
$^{1}$H-NMR (600 MHz),$^{13}$C-NMR (150 MHz) spectral data for compounds **3**–**5** in $CD_3OD$.

| position | 3 | | 4 | | 5 | |
|---|---|---|---|---|---|---|
| | $\delta_C$ | $\delta_H$ ($J$ in Hz) | $\delta_C$ | $\delta_H$ ($J$ in Hz) | $\delta_C$ | $\delta_H$ ($J$ in Hz) |
| 2 | 76.3 | 5.65 dd (12.8, 2.2) | 75.7 | 5.53 d (13.2) | 75.9 | 5.56 dd (12.5, 3.0) |
| 3 | 43.4 | 2.91 dd (18.6, 12.8) | 43.1 | 2.94 m | 43.0 | 2.98 dd (17.1, 12.5) |
| | | 2.78 dd (18.6, 2.2) | | 2.65 d (17.1) | | 2.64 dd (17.1, 3.0) |
| 4 | 198.2 | | 198.3 | | 198.3 | |
| 4a | 103.3 | | 103.3 | | 103.4 | |
| 5 | 165.5 | | 165.5 | | 165.5 | |
| 6 | 96.2 | 5.88 d (2.1) | 97.0 | 5.81 d (1.7) | 96.9 | 5.81 d (1.9) |
| 7 | 168.4 | | 168.2 | | 168.3 | |
| 8 | 97.1 | 5.82 d (2.1) | 96.2 | 5.85 d (1.7) | 96.1 | 5.85 d (1.9) |
| 8a | 165.2 | | 165.3 | | 165.3 | |
| 1′ | 123.7 | | 119.1 | | 117.7 | |
| 2′ | 153.5 | | 156.5 | | 154.7 | |
| 3′ | 98.4 | 6.86 s | 104.1 | 6.18 s | 99.4 | 6.35 s |
| 4′ | 156.9 | | 155.4 | | 159.4 | |
| 5′ | 121.2 | | 115.1 | | 122.0 | |
| 6′ | 119.7 | 7.60 d (2.6) | 125.8 | 7.01 s | 128.4 | 7.04 s |
| 1″ | | | 123.0 | 6.23 d (9.5) | 28.8 | 3.12 d (7.2) |
| 2″ | 145.3 | 7.50 d (2.2) | 128.7 | 5.43 d (9.5) | 124.3 | 5.15 t (7.2) |
| 3″ | 107.5 | 6.65 d (2.2) | 77.4 | | 132.6 | |
| 4″ | | | 28.3 | 1.30 s | 26.0 | 1.64 s |
| 5″ | | | 28.2 | 1.30 s | 17.8 | 1.61 s |
| OMe-4′ | | | | | 55.8 | 3.71 s |

(**1**–**5**) and paratrimerin F (**6**) (Fig. 5). Starting from umbelliferone, the biogenetic pathway of **1** and **6** involved prenylation, oxidation, and cyclization. And norartocarpetin under the oxidation, cyclization, methylation, and psoralen synthase catalyzed dealkylation to form compounds **2**–**5**. The prenylation of umbelliferone and norartocarpetin, was biosynthesized from **6** by Orf2 in the presence of $Mg^{2+}$ (Kuzuyama et al., 2005). NADPH oxidase generates superoxide by transferring electrons from NADPH to molecular oxygen to produce the superoxide anion, subsequently causing the monooxygenase reaction (Paravicini and Touyz, 2008). The methylation reaction was catalyzed by methyltransferase with *S*-adenosyl methionine (SAM) as a co-substrate (Struck et al., 2012). The oxidative bond cleavage of **4** that leads to **3** was catalyzed by cytochrome P450 requiring $O_2$ and NADPH as a co-factor (Stanjek et al., 1999).

Macrophages play a crucial role in the process of inflammation, which can be activated by LPS or cytokines, inducing the release of numerous pro-inflammatory mediators, such as NO, tumor necrosis factor-α (TNF-α), and so on. Excessive production of NO will activate the NF-κB signaling pathway and induce the production of pro-inflammatory factors, which in turn can activate iNOS and promote the body to produce more NO, thus forming a positive feedback loop to make inflammation more durable. Therefore, LPS induced RAW 264.7 cells were used to establish an inflammatory injury model, and NO was used as an indicator to evaluate the anti-inflammatory effect of the tested compounds. To determine if compounds **1**–**46** can modulate the NO production by macrophages, the Griess reagent was used to analyze the NO levels. The research results show that some compounds exhibited obvious NO production inhibitory activity as shown in Table 4. Among them, compound **27** was the most effective with $IC_{50}$ values of 15.6 ± 0.6 μM. Besides, the growth inhibitory effect of these compounds on RAW 264.7 cells was evaluated by the MTT assay. Cells were incubated with compounds at the concentrations (50 μM) after being pretreated with LPS or medium only. The result showed that cell viabilities were not significantly affected within the locatable ambit of concentration.

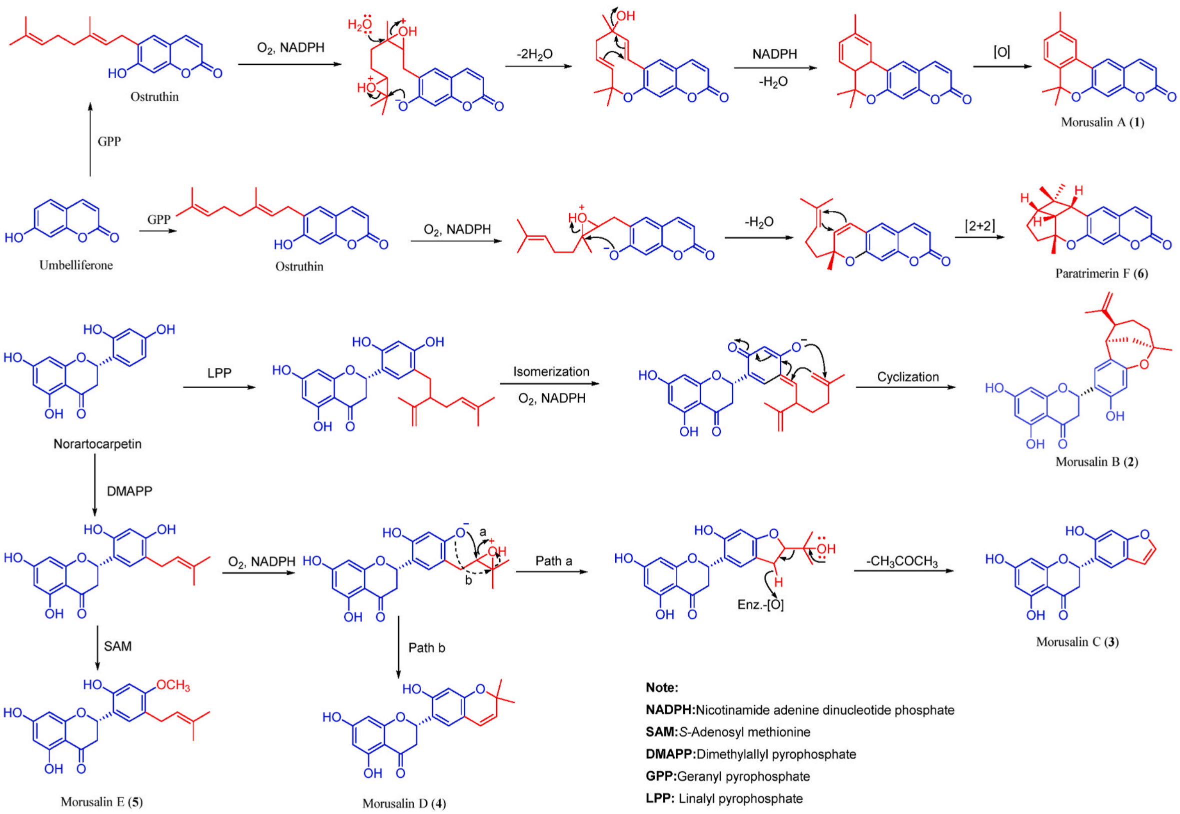


**Fig. 5.** Proposed biogenetic pathway for the formation of the compounds **1**–**6**.

**Table 4**
Inhibitory effects of compounds on LPS-induced NO production in macrophages ($IC_{50}$ values less than 50 μM are listed).

| Compounds | $CC_{50}$ (μM) | $IC_{50}$ (μM) |
|---|---|---|
| **6** | >50 | 45.1 ± 0.7 |
| **21** | >50 | 35.3 ± 1.2 |
| **24** | >50 | 30.1 ± 0.5 |
| **27** | >50 | 15.6 ± 0.6 |
| L-NMMA | >50 | 15.2 ± 0.9 |

The inhibitory rate of HCT 8 and HepG 2 cells growth of compounds **1**–**46** was measured by the cytotoxicity assay (MTT assay) and compared with a negative control and positive control (cisplatin, $IC_{50} = 16.7 \pm 0.9$ and 25.1 ± 1.9 μM, respectively). The results (Table 5) showed that compounds **11**, **12**, **19**, **22**, **23**, **31**, **35**, **36**, **37**, and **38** showed cytotoxicity against both human tumor cells, and compound **19** showed the strongest inhibitory effect with $IC_{50}$ of 12.5 ± 0.5 and 25.7 ± 2.1 μM, respectively.

**Table 5**
Cytotoxicity of compounds on HCT 8 and HepG 2 cell lines ($IC_{50}$ values less than 50 μM are listed).

| Compounds | $IC_{50}$ (μM) | | Compounds | $IC_{50}$ (μM) | |
|---|---|---|---|---|---|
| | HCT 8 | HepG 2 | | HCT 8 | HepG 2 |
| **8** | 29.5 ± 1.3 | >50 | **28** | 35.4 ± 0.7 | >50 |
| **9** | 36.3 ± 1.0 | >50 | **31** | 38.9 ± 1.8 | 37.1 ± 0.6 |
| **11** | 28.1 ± 0.8 | 26.9 ± 1.1 | **35** | 35.4 ± 2.1 | 35.4 ± 1.8 |
| **12** | 32.3 ± 1.7 | 28.8 ± 1.3 | **36** | 31.6 ± 1.9 | 37.1 ± 1.4 |
| **13** | 45.7 ± 1.5 | >50 | **37** | 28.8 ± 0.8 | 28.8 ± 0.9 |
| **19** | 12.5 ± 0.5 | 25.7 ± 2.1 | **38** | 25.1 ± 1.6 | 31.6 ± 2.3 |
| **22** | 31.9 ± 1.1 | 39.8 ± 1.8 | **42** | 33.1 ± 1.3 | >50 |
| **23** | 30.9 ± 1.9 | 32.3 ± 0.4 | **43** | 34.6 ± 1.1 | >50 |
| **24** | 36.3 ± 1.6 | >50 | cisplatin | 16.7 ± 0.9 | 25.1 ± 1.9 |

## 3. Conclusions

In this study, five previously undescribed compounds (**1**–**5**) and forty-one known ones (**6**–**46**) were isolated and identified from the roots bark of *M. alba*. Their structures with absolute configurations were elucidated by detailed interpretation of NMR, MS, single-crystal X-ray diffraction, and ECD calculations. The anti-inflammatory activity of all compounds was assessed by measuring NO production on LPS-induced RAW 264.7 cells, and some compounds exhibited significant inhibitory activity. Furthermore, the antiproliferative activities against the HCT 8 and HepG 2 cell lines of all isolates were assessed using the MTT method, and the results demonstrated that some compounds exhibited obvious cytotoxic activity. This study lays the foundation for studying the potential therapeutic applications of phenolic acids from *M. alba* for inflammatory diseases and hepatoma.

## 4. Experimental section

### 4.1. General experimental procedures

ESIMS and HRESIMS were performed by API-Qstar-TOF instrument (Applied Bio-system Corporation, Canada) or LC-IT-TOF instrument (AB-MDS Sciex, Concord, ON, Canada). NMR spectra were obtained using a Bruker AV 600 MHz spectrometer with tetramethylsilane as an internal standard (Bruker BioSpin Group, Germany). Optical rotations were determined by a Jasco DIP- 370 digital polarimeter (JASCO Corporation, Tokyo, Japan). IR spectra were obtained by using Vertex 70 FT-IR spectrophotometer (Bruker Optics, Germany). UV spectra were measured on a Shimadzu UV-2401A spectrophotometer (Shimadzu Corporation, Tokyo, Japan). ECD spectra of samples disproved in MeOH were recorded by V100 Chirascan instrument (cells: 1 mm, time-per-point: 1 s/nm (25 μs × 40,000, 195–400 nm), Applied Photophysics Limited, England). Semipreparative HPLC was conducted on an Agilent 1200 liquid chroma-tography (Zorbax SB-C18 column, 5 μm, 9.4 mm × 250 mm, Agilent, USA). Column chromatography (CC) was carried out on silica gel (200–300 mesh, Qingdao Marine Chemical Co. Ltd., P. R. China), MCI gel (CHP 20P, 75–150 μm, Mitsubishi Chemical Corporation, Tokyo, Japan), ODS-C18 (75 μm, YMC Co., Ltd., Japan) and Sephadex LH-20 (Amersham Biosciences AB, Uppsala, Sweden). Fractions were monitored by TLC plates (silica gel GF254, Qingdao Marine Chemical Co. Ltd., P. R. China). Visualization of spots was under UV light (254 nm) or by spraying with 5% $H_2SO_4$–EtOH followed by heating. Melting points were recorded on an SGM X-5 micro melting point apparatus.

### 4.2. Plant material

*M. alba* was collected from Chuxiong City, Yunnan Province in July 2021. After identification by Professor Xuan-Qin Chen of Kunming University of Science and Technology, the specimen (KUST20210609) is now stored in the Key Laboratory of Resource Pharmaceutical Chemistry of Kunming University of Science and Technology.

### 4.3. Extraction and isolation

The dried crushed tuber of *M. alba* (10 kg) was extracted with 75% ethanol (EtOH) three times, each 100 L, for 48 h, at room temperature to yield the crude EtOH extract. Subsequently, the EtOH fraction was suspended in water and mixed with ethyl acetate (EtOAc) (3 × 25 L) to obtain the EtOAc extract (1.4 kg). The ethyl acetate fraction was eluted by silica gel column ($CH_2Cl_2$/MeOH, 100:1 to 5:1, v/v) to yield eight fractions (Fr.1–Fr.8).

Fr.1 (16.5 g) utilized reversed-phase column chromatography, eluting with a gradient of MeOH/$H_2O$ (50:50 to 100:0, v/v) obtain ten subfractions (Fr.1-1–1-10). Fr.1-1 (360.9 mg) was subjected to silica gel CC and eluted isocratically with a petroleum ether-chloroform solvent system (1:3, v/v) to obtain compounds **28** (3.2 mg) and **27** (5.3 mg). Compounds **6** (2.9 mg) and **19** (3.7 mg) were obtained from Fr.1-2 (1.1 g) by silica gel CC (petroleum ether/acetone = 10:1, v/v). Fr.1-3 (520.3 mg) was separated over a silica gel CC ($CH_2Cl_2$/MeOH, 30:1 to 10:1, v/v) to afford compounds **8** (15.4 mg) and **3** (3.4 mg). Fr.1-4 (573.9 mg) was subjected to passage over a silica gel CC ($CHCl_3$/MeOH, 30:1, v/v) and precipitated in chloroform solvent to obtain compounds **1** (2.4 mg) and **29** (6.5 mg).

Separation of Fr.2 (33.9 g) by reversed-phase column chromatography, eluting with a gradient of (MeOH/$H_2O$, 50:50 to 100:0, v/v) to provide six sub-fractions (Fr.2-1–Fr.2-9). Fr.2-1 (952.6 mg) was selected for Sephadex LH-20 gel chromatography, eluted with MeOH, further purified by semi-preparative HPLC (MeOH/$H_2O$, 83:17, v/v, 3 mL/min) to obtain compounds **4** (5.7 mg) and **41** (4.6 mg). Compounds **42** (7.1 mg) and **7** (6.4 mg) were obtained from Fr.2-2 (721.2 mg) using gel column chromatography. Fr.2-3 (2.5 g) was chromatographed over a column of Sephadex LH-20 with (MeOH) to obtain compounds **24** (8.9 mg) and **11** (10.1 mg). Fr.2-4 (1.7 g) was applied to a Sephadex LH-20 column and eluted with MeOH to yield compounds **12** (9.8 mg), **25** (22.8 mg) and **43**. Fr.2-6 (676.4 mg) was subjected to passage over a Sephadex LH-20 column (MeOH) to give compounds **33** (3.2 mg) and **5** (3.1 mg). Fr.2-7 (1.5 g) was first purified by Sephadex LH-20 column (MeOH), and then further purified by silica gel column (PE/acetone, 8:1, v/v) to obtain compounds **21** (9.3 mg), **34** (5.5 mg) and **13** (7.6 mg). Fr.2-7-3 (330.7 mg) was by subjecting to a silica gel CC and eluted with ($CH_2Cl_2$/isopropanol, 20:1, v/v) to obtain compounds **18** (8.7 mg) and **2** (4.3 mg). Fr.2-8-1 (3.8 g) was separated on a silica gel CC ($CH_2Cl_2$/isopropanol, 20:1, v/v) and further purified twice on a Sephadex LH-20 column (MeOH) to give compounds **22** (11.3 mg) and **15** (8.5 mg). Fr.2-8-2 (879.2 mg) was separated on a Sephadex LH-20 column ($CHCl_3$/MeOH, 1:1, v/v) to give compounds **35** (9.0 mg) and **9** (7.7 mg). Fr.2-9-1 (2.2 g) was purified by Sephadex LH-20 column ($CHCl_3$/MeOH, 1:1, v/v) and silica gel CC (PE/acetone, 6:1, v/v) to obtain compounds **16** (8.6 mg) and **10** (10.0 mg). Fr.2-9-2 (923.3 mg) was applied to Sephadex LH-20 gel chromatography using ($CHCl_3$/MeOH, 1:1, v/v) as the eluent and purified by silica gel CC (PE/acetone, 5:1, v/v) to yield compounds **17** (11.2 mg) and **36** (9.5 mg).

Fr.3 (27.8 g) was separated into five fractions (Fr.3-1–Fr.3-5) on a reversed-phase column chromatography (MeOH/$H_2O$, 60:40 to 100:0, v/v). Fr.3-1 (2.3 g) was purified by Sephadex LH-20 column (MeOH) to obtain compounds **14** (4.0 mg) and **20** (10.0 mg). Fr.3-2 (1.3 g) was using silica gel CC eluted with $CH_2Cl_2$–MeOH (25:1, v/v) to get compounds **37** (80.0 mg) and **32** (12.2 mg). Compound **23** (11.3 mg) was isolated from Fr.3-1-3 (150.7 mg) using a silica gel column (PE/acetone, 5:1, v/v). Fr.3-2 (2.7 g) was separated by silica gel column ($CHCl_3$/isopropanol, 20:1, v/v) to obtain compounds **31** (10.6 mg) and **38** (20.8 mg). Fr.3-3 (1.2 g) was then applied to a silica gel column and eluted with ($CHCl_3$/isopropanol, 30:1, v/v) to yield compounds **30** (9.0 mg) and **39** (3.2 mg).

Fr.4 (39.7 g) was separated into seven fractions (Fr.4-1–Fr.4-7) by silica gel column ($CH_2Cl_2$/MeOH, 100:1 to 1:1, v/v), Fr.4-1 (3.8 g) was purified by silica gel column (PE/acetone, 5:1, v/v) and Sephadex LH-20 column (MeOH) to get compound **44** (12.8 mg). Compounds **45** (6.1 mg) and **46** (9.8 mg) were obtained from Fr. 4-2 (356.0 mg) by repeated silica gel CC (PE/acetone, gradient, 30:1 to 1:1), and then purified over a Sephadex LH-20 column ($CHCl_3$–MeOH, 1:1). Fr.4-3 (3.3 g) was separated by Sephadex LH-20 column (MeOH), and then purified by silica gel CC ($CH_2Cl_2$/isopropanol, 15:1, v/v) to obtain compounds **40** (9.8 mg) and **26** (8.0 mg).

### 4.4. Spectroscopic data

Morusalin A (**1**): white powder; $[\alpha]_D^{25}$ = - 0.67 (c 0.15, MeOH); UV (MeOH) $\lambda_{max}$ (log $\varepsilon$): 234 (3.50), 269 (3.40) nm; IR (KBr) $\nu_{max}$ 1733, 1624, 1456, 1150, 1120; $^1H$ (600 MHz, $CDCl_3$) and $^{13}C$ NMR (150 MHz, $CDCl_3$) data, see Table 1; HR-ESI-MS: *m/z* 293.1163 $[M+H]^+$ (calc. for $C_{19}H_{17}O_3^+$, 293.1172).

Morusalin B (**2**): colorless crystals in MeOH; mp 239.5−241.5 °C; $[\alpha]_D^{25}$ = - 0.67 (c 0.15, MeOH); UV(MeOH) $\lambda_{max}$ (log $\varepsilon$): 217 (3.87), 288 (3.77) nm; ECD (MeOH) $\lambda_{max}$ ($\Delta\varepsilon$) 211 (−14.17), 252 (0.63), 294 (−5.05) nm; IR (KBr) $\nu_{max}$ 3425, 1635, 1115, 1068; $^1H$ (600 MHz, $CD_3OD$) and $^{13}C$ NMR (150 MHz, $CD_3OD$) data, see Table 2; HR-ESI-MS: *m/z* 423.1813 $[M+H]^+$ (calc. for $C_{25}H_{27}O_6^+$, 423.1802).

Crystal data for morusalin B (**2**). $C_{25}H_{26}O_6$ (M = 422.47 g/mol): triclinic, space group P1, $a$ = 10.9041(4) Å, $b$ = 11.0715(4) Å, $c$ = 11.9136(5) Å, $V$ = 1181.24(8) Å$^3$, Z = 1, T = 100 K, μ (Mo Kα) = 0.093 mm$^{-1}$, $Dcalc$ = 1.264 g/cm$^3$, 5471 reflections measured (2.07° ≤ $\theta$ ≤ 27.60°), 5471 unique ($R_{int}$ = 0.0429, $R_{sig}$ = 0.0254) which were used in all calculations. The final $R_1$ was 0.1149 (I > 2$\sigma$(I)) and $wR_2$ was 0.1888 (all data). CCDC: 2358494 (https://www.ccdc.cam.ac.uk).

Morusalin C (**3**): white powder; $[\alpha]_D^{25}$ = - 72.5 (c 0.12, MeOH); UV (MeOH) $\lambda_{max}$ (log $\varepsilon$): 217 (3.90), 289 (3.80) nm; IR (KBr) $\nu_{max}$ 3390, 3228, 2959, 2925, 1640, 1158, 1091; $^1H$ (600 MHz, $CD_3OD$) and $^{13}C$ NMR (150 MHz, $CD_3OD$) data, see Table 3; HR-ESI-MS: *m/z* 313.0710 $[M+H]^+$ (calc. for $C_{17}H_{13}O_6^+$, 313.0707).

Morusalin D (**4**): white powder; $[\alpha]_D^{25}$ = - 26.7 (c 0.12, MeOH); UV (MeOH) $\lambda_{max}$ (log $\varepsilon$): 229 (5.96), 287 (3.80) nm; IR (KBr) $\nu_{max}$ 3432, 1638, 1160, 1088; $^1H$ (600 MHz, $CD_3OD$) and $^{13}C$ NMR (150 MHz, $CD_3OD$) data, see Table 3; HR-ESI-MS: *m/z* 355.1177 $[M+H]^+$ (calc. for $C_{20}H_{19}O_6^+$, 355.1176).

Morusalin E (**5**): white powder; $[\alpha]_D^{25}$ = - 29.6 (c 0.19, MeOH); UV (MeOH) $\lambda_{max}$ (log $\varepsilon$): 233 (3.81), 286 (3.72) nm; IR (KBr) $\nu_{max}$ 3412, 1639, 1159; $^1$H (600 MHz, $CD_3OD$) and $^{13}$C NMR (150 MHz, $CD_3OD$) data, see Table 3; HR-ESI-MS: *m/z* 371.1494 $[M+H]^+$ (calc. for $C_{21}H_{232}O_6^+$, 371.1489).

### 4.5. ECD calculation

Based on the ROESY spectra, Chem3D modeling, appropriate conformers of model compounds **2** was generated within 10 kcal/mol energy window under CONFLEX8/MMFF94s (Fujihara et al., 2005). The highest distribution of each conformer owning a relative energy within 10 kcal/mol was optimized using density functional theory (DFT) method at B3LYP/6-31G (d) level with the IEFP CM solvent model in MeOH by Gaussian 16 package of programs. The ECD calculations of optimized conformers were subsequently carried out with DFT-B3LYP/6–311 + G (2 d, p) level under the same solvent model (Pescitelli and Bruhn, 2016). The computational ECD spectra were generated via SpecDis 1.71 (Bruhn et al., 2013) according to the Boltzmann weighting and UV shift.

### 4.6. Cell viability assay

To measure cell viability, MTT assay was performed. RAW 264.7 cells (4 × 104 cells/well) were cultured in 96 well plates for 1 day, then were treated with several concentrations of compounds for 1 h, and then co-stimulated with 1 μg/mL LPS for another 24 h. After washing with PBS buffer twice, 20 μL of MTT solution (5 mg/mL in phosphate-buffered saline (PBS), pH 7.4) was added to each well and incubated for an additional 4 h. MTT was removed carefully and resolved with 150 μL/well DMSO. The optical density was measured at 490 nm using a Thermo Fisher Scientific microplate reader (Massachusetts, USA).

### 4.7. Anti-inflammatory activity assay

The anti-inflammatory activity was determined according to the method (Cao et al., 2016), and a specific inhibitor of NOS (L-NMMA) was used as a positive control. The inhibitory rate of NO production in RAW 264.7 cells was determined by Griess method to evaluate the anti-inflammatory activity of the compounds.

### 4.8. Cytotoxicity assay

Cytotoxic activity assays against HepG 2, HCT 8 cell lines were carried out using the method described previously (Khiev et al., 2012). The concentration of 50 μM was used as the initial screening concentration, and then five gradient concentrations were set for rescreening, each group was repeated three times. After 4 h of culture with MTT, the absorbance value was measured at 490 nm of the microplate reader, and the $IC_{50}$ value was calculated by Statistical Package for the Social Science (SPSS 21.0).

## CRediT authorship contribution statement

**Min Tan:** Writing – original draft, Investigation, Data curation. **Xiao-Cong Liu:** Methodology, Formal analysis. **Dan Liu:** Methodology, Formal analysis. **Xuan-Qin Chen:** Resources, Investigation. **Rong-Tao Li:** Visualization, Methodology. **Zhi-Jun Zhang:** Writing – review & editing, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.phytochem.2024.114352.

## Data availability

Data will be made available on request.

## References

Bruhn, T., Schaumloffel, A., Hemberger, Y., Bringmann, G., 2013. SpecDis: quantifying the comparison of calculated and experimental electronic circular dichroism spectra. Chirality 25, 243–249. https://doi.org/10.1002/chir.22138.

Cao, Y.G., Zheng, X.K., Yang, F.F., Li, F., Qi, M., Zhang, Y.L., Feng, W.S., 2018. Two new phenolic constituents from the root bark of *Morus alba* L. and their cardioprotective activity. Nat. Prod. Res. 32, 391–398. https://doi.org/10.1080/14786419.2017.1309535.

Cao, L., Li, R.T., Chen, X.Q., Xue, Y., Liu, D., 2016. Neougonin a inhibits lipopolysaccharide-induced inflammatory responses via downregulation of the NF-kB signaling pathway in RAW 264.7 macrophages. Inflammation 39, 1939–1948. https://doi.org/10.1007/s10753-016-0429-9.

Cazal, C.D.M., Domingues, V.D.C., Batalhão, J.R., Bueno, O.C., Edson, R.F., da Silva, M.F.D.G.F., Vieira, P.C., Fernandes, J.B., 2009. Isolation of xanthyletin, an inhibitor of ants' symbiotic fungus, by high-speed counter-current chromatography. J. Chromatogr., A 1216, 4307–4312. https://doi.org/10.1016/j.chroma.2009.02.066.

Chan, E.C., Wong, S.K., Lye, P.Y., Wong, S.K., 2016. Phytochemistry, pharmacology, and clinical trials of *Morus alba*. Chin. J. Nat. Med. 14, 17–30. https://doi.org/10.3724/sp.j.1009.2016.00017.

Chang, Y.S., Jin, H.G., Lee, H., Lee, D.S., Woo, E.R., 2019. Phytochemical constituents of the root bark from *Morus alba* and their IL-6 inhibitory activity. Nat. Prod. Science 25, 268–274. https://doi.org/10.20307/nps.2019.25.3.268.

Chen, L.Y., Duan, Y.H., Li, C., Wang, Y., Tong, X.P., Dai, Y., Yao, X.S., 2013. Four new prenylated flavonoids from the roots of *Cudrania tricuspidata*. Magn. Reson. Chem. 51, 842–846. https://doi.org/10.1002/mrc.4020.

Christensen, L.P., Lam, J., 1989. Excelsaoctaphenol, a stilbene dimer from *Chlorophora excelsa*. Phytochemistry 28, 917–918. https://doi.org/10.1016/0031-9422(89)80142-6.

Dang, P.H., Le, T.H., Phan, P.K.T., Le, P.T.T., Nguyen, M.T.T., Nguyen, N.T., 2017. Two acridones and two coumarins from the roots of *Paramignya trimera*. Tetrahedron Lett. 58, 1553–1557. https://doi.org/10.1016/j.tetlet.2017.02.083.

Dat, N.T., Binh, P.T.X., Quynh, L.T.P., Van Minh, C., Huong, H.T., Lee, J.J., 2010. Cytotoxic prenylated flavonoids from *Morus alba*. Fitoterapia 81, 1224–1227. https://doi.org/10.1016/j.fitote.2010.08.006.

Dong, H.B., Yu, P., Long, B., Peng, T., He, Y.J., Xu, B., Liao, L., Lu, L., 2023. Total synthesis of kuwanons A and B and discovery of their antibacterial mechanism. J. Nat. Prod. 86, 2022–2030. https://doi.org/10.1021/acs.jnatprod.3c00466.

Fukai, T., Hano, Y., Hirakura, K., Nomura, T., Uzawa, J., Fukushima, K., 1985. Structures of two natural hypotensive Diels-Alder Type adducts, mulberrofurans F and G, from the cultivated mulberry tree (*Morus ihou* KOIDZ.). Chem. Pharm. Bull. 33, 3195–3204. https://doi.org/10.1248/cpb.33.3195.

Fujihara, T., Obora, Y., Tokunaga, M., Sato, H., Tsuji, Y., 2005. Dendrimer N-heterocyclic carbene complexes with rhodium (I) at the core. Chem. Commun. 36, 4526–4528. https://doi.org/10.1039/B506927K.

Ha, M.T., Tran, M.H., Ah, K.J., Jo, K.J., Kim, J., Kim, W.D., Cheon, W.J., Woo, M.H., Ryu, S.H., Min, B.S., 2016. Potential pancreatic lipase inhibitory activity of phenolic constituents from the root bark of *Morus alba* L. Bioorg. Med. Chem. Lett. 26, 2788–2794. https://doi.org/10.1016/j.bmcl.2016.04.066.

Hao, J.Y., Gao, Y.F., Xue, J.B., Yang, Y.Y., Yin, J.J., Wu, T., Zhang, M., 2022. Phytochemicals, pharmacological effects and molecular mechanisms of Mulberry. Foods 11, 1170. https://doi.org/10.3390/foods11081170.

Hao, M.M., Huang, P.J., Ruan, J.Y., Sun, F., Han, L.F., Liu, M.Y., Zhang, Y., Wang, T., 2021. Bioactive flavonoids and stilbenes from the leaf of *Morus alba* var. *multicaulis*. Fitoterapia 154, 105018. https://doi.org/10.1016/j.fitote.2021.105018.

Jeong, S.H., Ryu, Y.B., Curtis-Long, M.J., Ryu, H.W., Baek, Y.S., Kang, J.E., Lee, W.S., Park, K.H., 2009. Tyrosinase inhibitory polyphenols from roots of *Morus ihou*. J. Agric. Food Chem. 57, 1195–1203. https://doi.org/10.1021/jf8033286.

Jung, J.W., Ko, W.M., Park, J.H., Seo, K.H., Oh, E.J., Lee, D.Y., Lee, D.S., Kim, Y.C., Lim, D.W., Han, D., 2015. Isoprenylated flavonoids from the root bark of *Morus alba* and their hepatoprotective and neuroprotective activities. Arch Pharm. Res. (Seoul) 38, 2066–2075. https://doi.org/10.1007/s12272-015-0613-8.

Kapche, G.D.W.F., Fozing, C.D., Donfack, J.H., Fotso, G.W., Amadou, D., Tchana, A.N., Bezabih, M., Moundipa, P.F., Ngadjui, B.T., Abegaz, B.M., 2009. Prenylated arylbenzofuran derivatives from *Morus mesozygia* with antioxidant activity. Phytochemistry 70, 216–221. https://doi.org/10.1016/j.phytochem.2008.12.014.

Khiev, P., Kim, J.W., Sung, S.J., Song, H.H., Chin, Y.W., Lee, H.K., Oh, S.R., 2012. Ingenane-type diterpenes with a modulatory effect on IFN-γ production from the roots of *Euphorbia kansui*. Arch Pharm. Res. (Seoul) 35, 1553–1558. https://doi.org/10.1007/s12272-012-0905-1.

Kim, S.B., Chang, B.Y., Hwang, B.Y., Kim, S.Y., Lee, M.K., 2014. Pyrrole alkaloids from the fruits of *Morus alba*. Bioorg. Med. Chem. Lett. 24, 5656–5659. https://doi.org/10.1016/j.bmcl.2014.10.073.

Kuzuyama, T., Noel, J.P., Richard, S.B., 2005. Structural basis for the promiscuous biosynthetic prenylation of aromatic natural products. Nature 435, 983–987. https://doi.org/10.1038/nature03668.

Lan, W.C., Tzeng, C.W., Lin, C.C., Yen, F.L., Ko, H.H., 2013. Prenylated flavonoids from *Artocarpus altilis*: antioxidant activities and inhibitory effects on melanin production. Phytochemistry 89, 78–88. https://doi.org/10.1016/j.phytochem.2013.01.011.

Langeder, J., Doering, K., Schmietendorf, H., Grienke, U., Schmidtke, M., Rollinger, J.M., 2023. $^{1}$H NMR-based biochemometric analysis of *Morus alba* extracts toward a multipotent merbal anti-Infective. J. Nat. Prod. 86, 8–17. https://doi.org/10.1021/acs.jnatprod.2c00481.

Lin, Q.H., Yuan, J.B., Ma, Z.L., Liang, J., Zhai, X.X., Khan, I.A., Ding, Y.Q., Ren, G.L., Qin, H., 2016. Isoprenylated flavonoids from roots of *Artocarpus styracifolius*. Nat. Prod. Commun. 11, 1934578X1601101217. https://doi.org/10.1177/1934578X1601101217.

Luo, G.Y., Yang, Y., Zhou, M., Ye, Q., Liu, Y., Gu, J., Zhang, G.L., Luo, Y.G., 2014. Novel 2-arylbenzofuran dimers and polyisoprenylated flavanones from *Sophora tonkinensis*. Fitoterapia 99, 21–27. https://doi.org/10.1016/j.fitote.2014.08.019.

Ma, M.R., Zhou, T.X., Zhao, P., Choi, H.Y., Hao, J., Huang, H.Q., Wu, C.Q., Yang, X.Z., Pang, K.J., 2021. New flavonoids from the roots of *Sophora davidii* (Franch.) Skeels and their glucose transporter 4 translocation activities. Bioorg. Chem. 106, 104500. https://doi.org/10.1016/j.bioorg.2020.104500.

Paravicini, T.M., Touyz, R.M., 2008. NADPH oxidases, reactive oxygen species, and hypertension : clinical implications and therapeutic possibilities. Diabetes Care 31, S170–S180. https://doi.org/10.2337/dc08-s247.

Paudel, P., Seong, S.H., Zhou, Y., Ha, M.T., Min, B.S., Jung, H.A., Choi, J.S., 2019. Arylbenzofurans from the root bark of *Morus alba* as triple inhibitors of cholinesterase, β-Site amyloid precursor protein cleaving enzyme 1, and glycogen synthase kinase-3β: relevance to Alzheimer's Disease. ACS Omega 4, 6283–6294. https://doi.org/10.1021/acsomega.9b00198.

Pescitelli, G., Bruhn, T., 2016. Good computational practice in the assignment of absolute configurations by TDDFT calculations of ECD spectra. Chirality 28, 466–474. https://doi.org/10.1002/chir.22600.

Pu, H.Y., Cao, D.Y., Zhou, X., Li, F., Wang, L., Wang, M.K., 2024. Isoprenylated flavonoids and 2-arylbenzofurans from the root bark of *Morus alba* L. and their cytotoxic activity against HGC 27 cancer cells. Molecules 29, 30. https://doi.org/10.3390/molecules29010030.

Ryu, Y.B., Ha, T.J., Curtis-Long, M.J., Ryu, H.W., Gal, S.W., Park, K.H., 2008. Inhibitory effects on mushroom tyrosinase by flavones from the stem barks of *Morus lhou* (S.) Koidz. J. Enzym. Inhib. Med. Chem. 23, 922–930. https://doi.org/10.1080/14756360701810207.

Shi, Y.Q., Fukai, T., Sakagami, H., Chang, W.J., Yang, P.Q., Wang, F.P., Nomura, T., 2001. Cytotoxic flavonoids with isoprenoid groups from *Morus mongolica*. J. Nat. Prod. 64, 181–188. https://doi.org/10.1021/np000317c.

Shi, Y.Q., Nomura, T., Fukai, T., 2007. A new 2-arylbenzofuran from the root bark of Chinese *Morus cathayana*. Fitoterapia 78, 617–618. https://doi.org/10.1016/j.fitote.2007.06.009.

Shim, S.Y., Sung, S.H., Lee, M., 2018. Anti-inflammatory activity of mulberrofuran K isolated from the bark of *Morus bombycis*. Int. Immunopharm. 58, 117–124. https://doi.org/10.1016/j.intimp.2017.11.002.

Slade, D., Ferreira, D., Marais, J.P.J., 2005. Circular dichroism, a powerful tool for the assessment of absolute configuration of flavonoids. Phytochemistry 66, 2177–2215. https://doi.org/10.1016/j.phytochem.2005.02.002.

Smejkal, K., Svacinova, J., Slapetova, T., Schneiderova, K., Dall'Acqua, S., Innocenti, G., Zavalova, V., Kollar, P., Chudik, S., Marek, R., 2010. Cytotoxic activities of several geranyl-substituted flavanones. J. Nat. Prod. 73, 568–572. https://doi.org/10.1021/np900681y.

Stanjek, V., Miksch, M., Lueer, P., Matern, U., Boland, W., 1999. Biosynthesis of psoralen: mechanism of a cytochrome P450 catalyzed oxidative bond cleavage. Angew. Chem., Int. Ed. 38, 400–402. https://doi.org/10.1002/(sici)1521-3773(19990201)38:3<400::aid-anie400>3.0.co;2-k.

Struck, A.W., Thompson, M.L., Wong, L.S., Micklefield, J., 2012. S-Adenosyl-methionine-dependent methyltransferases: highly versatile enzymes in biocatalysis, biosynthesis and other biotechnological applications. Chembiochem 13, 2642–2655. https://doi.org/10.1002/cbic.201200556.

Takasugi, M., Nagao, S., Masamune, T., Shirata, A., Takahashi, K., 1979. Structures of moracins E, F, G, and H, new phytoalexins from diseased mulberry. Tetrahedron Lett. 20, 4675–4678. https://doi.org/10.1016/S0040-4039(01)86680-6.

Tseng, T.H., Chuang, S.K., Hu, C.C., Chang, C.F., Huang, Y.C., Lin, C.W., Lee, Y.J., 2010. The synthesis of morusin as a potent antitumor agent. Tetrahedron 66, 1335–1340. https://doi.org/10.1016/j.tet.2009.12.002.

Weng, M.S., Liao, C.H., Chen, C.N., Wu, C.L., Lin, J.K., 2007. Propolin H from taiwanese propolis induces G1 arrest in human lung carcinoma cells. J. Agric. Food Chem. 55, 5289–5298. https://doi.org/10.1021/jf070201n.

Xia, C.L., Tang, G.H., Guo, Y.Q., Xu, Y.K., Huang, Z.S., Yin, S., 2019. Mulberry Diels-Alder-type adducts from *Morus alba* as multi-targeted agents for Alzheimer's disease. Phytochemistry 157, 82–91. https://doi.org/10.1016/j.phytochem.2018.10.028.

Yang, Z.Z., Wang, Y.C., Wang, Y., Zhang, Y.F., 2012. Bioassay-guided screening and isolation of α-glucosidase and tyrosinase inhibitors from leaves of *Morus alba*. Food Chem. 131, 617–625. https://doi.org/10.1016/j.foodchem.2011.09.040.

Yang, Z.G., Matsuzaki, K., Takamatsu, S., Kitanaka, S., 2011. Inhibitory effects of constituents from *Morus alba* var. *multicaulis* on differentiation of 3T3-L1 cells and nitric oxide production in RAW264.7 cells. Molecules 16, 6010–6022. https://doi.org/10.3390/molecules16076010.

Yang, Y.N., Zhu, H., Yuan, X., Zhang, X., Feng, Z.M., Jiang, J.S., Zhang, P.C., 2021. Seven new prenylated flavanones from the roots of *Sophora flavescens* and their anti-proliferative activities. Bioorg. Chem. 109, 104716. https://doi.org/10.1016/j.bioorg.2021.104716.

Zhang, Y.L., Luo, J.G., Wan, C.X., Zhou, Z.B., Kong, L.Y., 2014. Geranylated 2-arylbenzofurans from *Morus alba* var. *tatarica* and their α-glucosidase and protein tyrosine phosphatase 1B inhibitory activities. Fitoterapia 92, 116–126. https://doi.org/10.1016/j.fitote.2013.10.017.

Zheng, X.K., Cao, Y.G., Ke, Y.Y., Zhang, Y.L., Li, F., Gong, J.H., Zhao, X., Kuang, H.X., Feng, W.S., 2017. Phenolic constituents from the root bark of *Morus alba* L. and their cardioprotective activity *in vitro*. Phytochemistry 135, 128–134. https://doi.org/10.1016/j.phytochem.2016.12.006.

Zheng, Z.P., Tan, H.Y., Wang, M.F., 2012. Tyrosinase inhibition constituents from the roots of *Morus australis*. Fitoterapia 83, 1008–1013. https://doi.org/10.1016/j.fitote.2012.06.001.

Zoofishan, Z., Kusz, N., Csorba, A., Toth, G., Hajagos-Toth, J., Kothencz, A., Gaspar, R., Hunyadi, A., 2019. Antispasmodic activity of prenylated phenolic compounds from the root bark of *Morus nigra*. Molecules 24, 2497. https://doi.org/10.3390/molecules24132497.

Zuo, G.Y., Yang, C.X., Han, J., Li, Y.Q., Wang, G.C., 2018. Synergism of prenylflavonoids from *Morus alba* root bark against clinical MRSA isolates. Phytomedicine 39, 93–99. https://doi.org/10.1016/j.phymed.2017.12.023.

Zuo, G.Y., Yang, C.X., Ruan, Z.J., Han, J., Wang, G.C., 2019. Potent anti-MRSA activity and synergism with aminoglycosides by flavonoid derivatives from the root barks of *Morus alba*, a traditional Chinese medicine. Med. Chem. Res. 28, 1547–1556. https://doi.org/10.1007/s00044-019-02393-7.